

Sleep Phenomics Automation (SPA): A Web Tool for Sleep Signal Phenotype Extraction

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Abstract

Purpose: Sleep contains rich multi-organ physiology, which can be accessed through polysomnography (PSG). Yet, the quantification of a comprehensive set of phenotypes, which we call sleep phenomics, poses a computational barrier for clinicians and researchers interested in education, exploratory analysis, and hypothesis-driven research. Meanwhile, the rapid development of artificial intelligence (AI) and various PSG markers proposed in the literature are further widening this barrier.

Methods: Here, we propose Sleep Phenomics Automation (SPA), a web-based platform that enables users to upload a sleep recording in the European Data

Format (EDF), visualize the signals, and download the phenomics in a tabular comma-separated values (CSV) format. The target users are sleep medicine trainees, clinicians, and researchers interested in sleep phenotypes or conducting research.

Results: We provided a description of how to use SPA, accompanied by screenshots. Using SPA, we demonstrated an age-dependent decrease in delta-band power during N3. The first version of SPA includes a small selected set of sleep phenotypes suitable for the adult population that cover both macrostructure (hypnogram and hypnodensity) and microstructures, including Complete AI Sleep Report, electroencephalography (EEG) spectrogram, EEG band Power, spindle, slow oscillation, EEG-based brain age, hypoxic burden, self-similarity, and heart rate variability. More phenotypes will be added in later versions. Advanced users proposing new phenotypes can experiment with them on the SPA website and contact SPA to make them available to the public.

Conclusion: SPA is for educational and research purposes. It requires zero coding for basic users and provides a centralized, web-based, no-installation interface, lowering the barrier to PSG phenotype extraction, visualization, and reproducible sleep research. SPA is available for non-commercial use at <https://sleep-phenomics-spa.ngrok.app>.

Keywords: Polysomnography, Sleep Phenotypes, Signal Analytics, Web Tool

Statements and Declarations

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1 Introduction

Polysomnography (PSG) is the clinical gold standard for diagnosing sleep disorders, generating rich multi-organ physiological data that includes electroencephalography (EEG), electromyography (EMG), electrooculography (EOG), electrocardiography (ECG), respiratory signals, and oxygen saturation [1, 2]. PSG signals contain diverse phenotypic information that can help reveal physiological processes relevant to neurodegenerative disorders, cardiovascular disease, metabolic dysfunction [3–6], etc. Recent progress made in signal processing and artificial intelligence (AI) for PSG signals has enabled the automation of sleep pattern extraction (i.e., feature extraction in AI terminology) extraction, sleep staging, and health outcome predictions [7–13]. However, the broader use of these research findings in education and research has been hindered by a critical gap: many researchers and clinicians who generate clinically relevant hypotheses often lack the computational expertise to quantify the phenotypes and test them, while the tools required to extract phenotypes remain inaccessible to non-programmers [14–16].

Several platforms have been developed to address the need for PSG analysis tools. Moonlight is an interactive viewer for polysomnographic (PSG) data built on top of the Luna software ecosystem [17]. It supports summary statistics, signal viewing, automated sleep staging, and comprehensive analysis of sleep signals. However, extracting each phenotype requires command-line input, which can hinder adoption by clinicians who lack specialized computational or informatics training. LightWAVE, available through PhysioNet [18], enables interactive visualization and basic annotation of signals from PhysioNet’s PhysioBank archives, but it does not offer automated phenotype extraction, limiting its utility as an analytical platform. Altamira, hosted on the National Sleep Research Resource (NSRR) [19, 20], offers a basic interactive viewer without analytical capabilities. YouSleep based on U-sleep [21] provides a web-based tool for high-frequency sleep staging, without phenotype extraction functionality. Other web-based tools, such as hypoxic burden calculator and Cerebra MY Sleep Scoring, have been discontinued. Local software solutions such as Visbrain [22] and Lunascope [17] require installation of the software and its dependent software, complicating accessibility in certain settings. Collectively, these platforms either require substantial technical expertise, lack the breadth of clinically relevant phenotypes, or function as simple signal viewers that are insufficient to support individual-level phenotype extraction, which is essential for their utility in education, exploratory analysis, and reproducible PSG-based research.

Here, we propose Sleep Phenomics Automation (SPA), a user-friendly web-based platform designed to democratize access to PSG-derived phenotypes, that addresses the limitations of existing tools. SPA provides a centralized, no-installation, upload-and-download interface for computing a comprehensive suite of sleep phenotypes (i.e., phenomics), including sleep architectural parameters, EEG microstructures, ECG patterns, and novel burden metrics. By eliminating the need for programming knowledge and offering immediate, downloadable results, SPA enables clinicians and researchers to explore PSG-derived phenotypes, support hypothesis generation, education, research, and better understand how sleep physiology is quantified. Furthermore,

SPA is also a platform for disseminating novel sleep phenotypes, ensuring that new phenotypes proposed in the literature are publicly accessible.

2 Methods

2.1 Website Design

SPA operates by allowing users to upload a sleep recording file in European Data Format (EDF) and then download extracted phenotypes in a tabular comma separated values (CSV) format. The web application utilizes a server-rendered architecture built on Flask with a glassmorphic UI design implemented through JavaScript and Cascading Style Sheets (CSS). Data visualization is primarily handled by Plotly.js for interactive plots. The backend uses MNE [23] for signal processing. An example sleep recording in EDF format and an example annotation file in CSV format are provided. SPA is licensed under CC BY-NC 4.0 for non-commercial use.

2.2 Implemented Phenotypes

For sleep stages and events, the user has the option to either upload annotation file or run automatic AI analysis, where the latter is implemented by Complete AI Sleep Report (CAISR) [24] for automatic sleep staging, arousal detection, apnea detection, and limb movement detection. The CAISR analytical components, including preprocessing, sleep staging, arousal detection, respiratory-event detection, limb-movement detection, and report generation, are packaged in task-specific Docker containers to isolate software dependencies and facilitate reproducible execution. These containers are supported on Linux and, on macOS, through Colima, with x86-64 emulation used for selected legacy CAISR environments on Apple Silicon. Apnea detection includes obstructive, central, and mixed apnea, and hypopnea. CAISR was developed using four diverse and large datasets including 25,749 participants and was externally validated in three datasets annotated by multiple experts. CAISR achieved non-inferiority performance compared to human experts based on inter-rater agreement in sleep staging, arousal detection, and limb movement detection in all external datasets, as well as non-inferiority performance in apnea detection in 2/3 external datasets. A hypnogram (30-second epochs) and a hypnodensity graph [25] are shown on the SPA webpage after generation. Common sleep architecture parameters (i.e., macrostructure) and apnea-hypopnea index (AHI), arousal index (ArI), hypoxic burden [26], arousal burden (AB) [27], and periodic limb movement index (PLMI) are downloadable. When CAISR is used for automatic apnea detection or the uploaded annotation file contains apnea subtypes, SPA computes obstructive AHI, central AHI, mixed AHI, and hypopnea index during rapid eye movement (REM) stage, Non-REM (NREM) stage, and total sleep time conditions.

For EEG-derived phenotypes, there are spectrogram (time-frequency representation) and average band power across sleep stages. The spectrogram is implemented by applying multitaper spectral analysis [28, 29] to 30-second epochs. The spectrogram is converted into decibel scale with a unit of $10 \log_{10}(\mu V^2/\text{Hz})$. A spectrogram

is shown on the SPA webpage after generation. The band powers include delta (1-4Hz), theta (4-8Hz), alpha (8-12Hz), sigma (11-15Hz), and beta (12-30Hz). Spindle and slow oscillation (SO) are detected using Luna [17]. The spindle phenotypes include density, frequency, duration, amplitude, and chirp (within-spindle frequency change rate). Spindles are detected at each EEG channel, and stratified by fast/slow/all as well as SO-coupled/uncoupled. SO phenotypes include density, amplitude, duration, and event-level rising slope (uV/s), defined from the negative peak to the next negative-to-positive zero crossing. SPA reports its mean across detected SOs. This is a local waveform-morphology measure, not overnight slow-wave activity decay across NREM cycles. Spindle-SO coupling phenotypes include overlap percentage and phase angles. For spindle detection quality control, the user can specify the quality cutoff (default 1) at the webpage that determines how much the detected spindles should stand out from the background. There is also EEG-based brain age (BA) [30] for adults. Brain age estimates the age from sleep EEG patterns. A higher brain age compared to the chronological age (CA), or a more positive brain age index (BAI = BA - CA) indicates accelerated brain aging and is associated with mortality [31] and future dementia at the population level [32].

For respiration-oximetry-derived phenotypes, there are hypoxic burden (HB) [26, 33] and self-similarity [34]. Here, we provided a custom implementation of HB according to the paper which quantifies the area under the desaturation curve for each desaturation event [33], combining both depth and duration of oxygen drops. These event-specific areas are then summed across the night and normalized to total sleep time to yield a burden value. HB measures the cumulative impact of oxygen desaturations during sleep, reflecting the physiological stress of recurrent hypoxemia on cardiovascular and metabolic systems. Self-similarity refers to the cyclical repetition of waxing-and-waning breathing patterns during sleep, reflecting high loop gain, i.e., instability in ventilatory control [34]. It is measured from respiratory inductance plethysmography (RIP) signals by detecting central apneas/hypopneas and quantifying the similarity of consecutive cycles using cross-correlation. A similarity index greater than 0.8 indicates strongly repeating cycles, and “full night similarity” is the fraction of total sleep time with such high similarity.

For ECG-derived phenotypes, we provided heart rate variability (HRV) [35] metrics. The HRV metrics cover time-domain, frequency-domain, and complexity-domain features, stratified by sleep stages. There are 25 metrics, including mean(nni), sd(nni), nni50, pnni50, nni20, pnni20, median(nni), range(nni), cv(nni), rmssd, sdsd, cvsd, mean(hr), max(hr), min(hr), std(hr), lf, hf, lfnu, hfnu, lf/hf ratio, vlf, sd1, sd2, and sd2/sd1 ratio as implemented in the `hrvanalysis` Python package.

To better inform the users, each phenotype is explained in a separate documentation webpage after clicking the question mark next to the phenotype name, with the following sections: Overview, Output Phenotypes, Clinical Implication (including actionable thresholds, if available), Strengths, Limitations, Calculation Method, and Reference.

2.3 Adding New Phenotypes

Advanced user can add new phenotypes, especially for authors who publish new sleep phenotypes and aim to make them accessible to clinicians and researchers. This function requires coding. SPA provides a unified Python function `get_custom_phenotype()` with a standard input and output arguments. The user can

first provide the implemented function in the text box on the SPA webpage. Once it can run on the example EDF file, the user can contact the maintainers of SPA to add it permanently. The details have been explained in the documentation webpage after clicking the question mark sign. An important direction in future versions is to add phenotypes from pediatric sleep studies.

2.4 Data Security and Privacy

Users are responsible for ensuring data privacy, including confirming that uploaded sleep recordings and annotation files have been de-identified, a function commonly provided in clinical PSG software. SPA displays a warning message before upload to remind the user of this requirement. Each analysis is performed within a unique, access-restricted job directory containing the uploaded recording and all generated intermediate and output files. File deletion is implemented within strict failure-handling logic and is performed after both successful and failed processing attempts, including parser, signal-processing, subprocess, cancellation, and timeout errors. Cleanup is completed before the analysis is reported as finished. SPA also performs a startup purge of abandoned upload and processing files older than the configured retention period, which is one hour by default, to address files remaining after abrupt server or machine termination. Cleanup operations are restricted to configured SPA storage directories and are logged without including uploaded filenames, physiological signals, annotations, or other protected health information.

3 Results

SPA is available at <https://sleep-phenomics-spa.ngrok.app/>. After logging in, users can upload signal files in the European Data Format (EDF), which can be exported by many PSG software. Then, the user is taken to a screen where they can map channels from their uploaded EDF file to a standardized list of channel names. If a user is missing a channel or wishes to exclude it, they can select “Does not exist”. After channel mapping, users are directed to a webpage where they can select the phenotypes to compute and see the signals in an interactive viewer (Figure 1). The users then click the “Compute Phenomics” button, and wait for the computation to complete. After completion, the “Compute Phenomics” button becomes “Download Phenomics”. Users can download sleep phenomics in tabular CSV format and the resulting figures are displayed on the webpage (Figure 2).

SPA allows adding new phenotypes (Figure 3), where we have provided detailed documentation with examples. However, we assume users who want to add new phenotypes have coding ability. Therefore, the typical users would be researchers who have developed and published new PSG-based markers, and who want to make their phenotype available to the public without coding burden.

To test SPA, an example EDF file was loaded (included in SPA), which has 204.2MB with 26 channels and 7.1 hours long. We ran all phenotypes on a machine with Intel(R) Xeon(R) W-2145 CPU 3.70GHz and 128GB of memory. The running time scale is shown in Table 1.

Using SPA, we created a scatterplot of age vs. the delta band power (1-4Hz) during N3 at C4-M1 channel in 50 random male participants from the Sleep Heart Health Study (SHHS) [36] hosted on National Sleep Research Resource (NSRR) [19, 20] in Figure 4. The participants were randomly sampled to span ages 40 to 80 years old. We observed a decreasing trend with increasing age, as expected from prior literature [37].

4 Discussion

SPA represents a practical step to a broader access to PSG-derived phenotypes. SPA is fully web-based and requires no coding or installation, which will lower the technical barrier that has limited the use of PSG-derived phenotypes by trainees, clinicians, and researchers. SPA is an educational and research-facing platform that provides access to state-of-the-art sleep phenotypes.

The current implementation of SPA does not and will not include all phenotypes, as new phenotypes are constantly proposed in the literature. New phenotypes are encouraged to be added. For example, PSG-based apnea endotyping holds promise to disentangle the heterogeneous nature of sleep apnea, where various PSG-based algorithms have been proposed [38–40]; Automated detection of cyclic alternating pattern (CAP) [41, 42] will help the clinician understand the patient’s sleep stability and fatigue symptoms [43]; EMG-derived phenotypes need to be added, such as sleep atonia index (SAI) [44, 45]. SAI reflects the degree of muscle atonia during REM. A reduced SAI is indicative of REM sleep behavior disorder (RBD), and can help identify patients at risk for Parkinson’s disease [46]. Importantly, cross-organ coupling metrics are encouraged. For example, Δ HR quantifies the heart rate response to respiratory events, reflecting autonomic nervous system activation and stress from sleep-disordered breathing [47]. We also encourage phenotypes targeted at the pediatric population, such as sleep staging [48] and brain age [49].

SPA can be used to help users explore and contextualize visual findings from PSG recordings. For example, an EEG alpha-delta finding [50] can be confirmed by the spectrogram; a benzodiazepine medication effect leading to excessive spindling can be confirmed by a strong sigma band in the spectrogram; deep desaturations associated with apneic events can be confirmed by elevated hypoxic burden; REM-dominant sleep apnea can be confirmed by a higher obstructive AHI than central AHI; and Cheyne-Stokes breathing can be confirmed by strong self-similarity metrics.

SPA can be used for hypothesis generation [51]. For example, a researcher may explore whether high respiratory self-similarity values suggest unstable ventilatory control patterns in a PSG recording. Follow-up recordings or external datasets could then be used to examine whether these patterns are reproducible across individuals or conditions. This approach supports endophenotyping and research hypothesis generation, while clinical interpretation or treatment decision is the downstream task.

SPA has several limitations. (1) SPA is not designed for batch processing, i.e., processing multiple sleep recordings at once, because the current implementation focuses on interactive single-recording exploration, visualization, and phenotype extraction. (2) While SPA automates feature extraction, the clinical interpretation of sleep phenotypes and the determination of actionable thresholds still requires domain expertise.

(3) SPA relies on the quality of EDF data provided, meaning that missing or noisy channels can limit the reliability of certain extracted phenotypes. (4) As a web-based platform, although SPA implements warnings for patient privacy data and immediate deletion after analysis, it may still require appropriate institutional approval and secure data handling.

Supplement

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Conflict of interest

MBW is a co-founder, scientific advisor and consultant to Beacon Biosignals and has a personal equity interest in the company. RJT reports: (1) patent and licence to MyCardio LLC, for ECG-spectrogram; (2) grant support, licence and intellectual property (licensed patents) from DeVilbiss-Drive Healthcare; (3) unlicensed patent for a device regulation CO₂ during positive pressure therapy, for central sleep apnea; (4) patent submitted for respiratory self-similarity for estimation of expressed high loop gain; (5) patent submitted for Enhanced Expiratory Rebreathing Space for treatment of high loop gain sleep apnea; (6) general sleep medicine consulting: GLG Councils, Guidepoint Global; (5) Consultant-Jazz Pharmaceuticals.

Data availability

SPA is available for non-commercial use at <https://sleep-phenomics-spa.ngrok.app/>.

Code availability

The SPA code is hosted at <https://github.com/bdsp-core/sleep-phenomics-automation>.

Author contribution

Conceptualization: HS, MBW, RJT; Data curation: MGA, SR, SW, NT, HS; Formal analysis: MGA, SR, SW; Funding acquisition: HS, MBW; Writing – original draft: MGA, SR, HS; Writing – review & editing: All authors.

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Tables

Table 1 Running time scale for phenotypes currently implemented in SPA for one example EDF file

Operation	Approximate Running Time
Sleep staging using CAISR	10 minutes
Spectrogram and band power	10 seconds
Spindle and slow oscillation	15 minutes
Brain age	2 minutes
Hypoxic burden	10 seconds
Self-similarity	2 minutes
Heart Rate Variability	1 minute
Cardiopulmonary Coupling	1 minute

Figure Captions

- Figure 1. Users can upload a de-identified EDF file or use the example EDF file. The signals are then displayed with full navigation functions. The available phenotypes are provided as check boxes, from which the users can choose.
- Figure 2. The resulting figures, such as hypnogram, hynodensity, and EEG spectrogram are shown on the webpage.
- Figure 3. SPA allows adding custom phenotype.
- Figure 4. Scatterplot of age vs. delta band power at C4-M1 channel during N3 in 50 random male participants from the Sleep Heart Health Study (SHHS).